Coding Challenge 5

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Contents

Question 1

Download two .csv files from Canvas called DiversityData.csv and Metadata.csv, and read them into R using relative file paths.

```
library("dplyr")
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library("tidyr")
library("ggplot2")
Metadata <- read.csv("Metadata.csv", na.strings="na")
DiversityData <- read.csv("DiversityData.csv", na.strings="na")
```

Question 2

Join the two dataframes together by the common column 'Code'. Name the resulting dataframe alpha.

```r
alpha<-inner_join(Metadata,DiversityData,by="Code")
str(alpha)
```

```
## 'data.frame':    70 obs. of  9 variables:
##  $ Code          : chr  "S01_13" "S02_16" "S03_19" "S04_22" ...
##  $ Crop          : chr  "Soil" "Soil" "Soil" "Soil" ...
##  $ Time_Point    : int  0 0 0 0 0 0 6 6 6 6 ...
##  $ Replicate     : int  1 2 3 4 5 6 1 2 3 4 ...
##  $ Water_Imbibed: num  NA NA NA NA NA NA NA NA NA NA ...
##  $ shannon       : num  6.62 6.61 6.66 6.66 6.61 ...
##  $ invsimpson    : num  211 207 213 205 200 ...
##  $ simpson       : num  0.995 0.995 0.995 0.995 0.995 ...
##  $ richness      : int  3319 3079 3935 3922 3196 3481 3250 3170 3657 3177 ...
```

Question 3

Calculate Pielou's evenness index: Pielou's evenness is an ecological parameter calculated by the Shannon diversity index (column Shannon) divided by the log of the richness column.

```r
alpha$log_Richness <- log(alpha$richness)
alpha_even <- mutate(alpha, Pielou_Evenness = (shannon/(alpha$log_Richness)))
```

Question 4

Using tidyverse language of functions and the pipe, use the summarise function and tell me the mean and standard error evenness grouped by crop over time.

```r
alpha_average <- alpha_even %>%
  group_by(Crop, Time_Point) %>%
  summarise(even.mean = mean(Pielou_Evenness),
            n = n(),
            sd.dev = sd(Pielou_Evenness)) %>%
  mutate(std.err = sd.dev/sqrt(n))
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by Crop and Time_Point.
## i Output is grouped by Crop.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(Crop, Time_Point))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

Question 5

Calculate the difference between the soybean column, the soil column, and the difference between the cotton column and the soil column

```r
alpha_average2 <- alpha_average %>%
  select(Time_Point,Crop,even.mean) %>%
  pivot_wider(names_from = Crop, values_from = even.mean) %>%
  mutate(diff.cotton.even = Soil-Cotton) %>%
  mutate(diff.soybean.even = Soil-Soybean)
```

Question 6

Connecting it to plots

```
select(alpha_average2, Time_Point,diff.cotton.even,diff.soybean.even) %>%
pivot_longer(c(diff.cotton.even,diff.soybean.even), names_to = "diff") %>%
ggplot(aes(x=Time_Point, y=value, color = diff)) +
  geom_line() +
  xlab("Time(hrs)") +
  ylab("Difference from soil in Pielou's Evenness")
```